Assignment 3 - Demon Slayer Corps: The String Hunt

Final Selection: test_matchers.py 24/24 PASS

Mission 1 - The Crow's Alarm (KMP)

```
pi = [0, 0, 1, 2, 3, 4]
Alarm count: 415
First 10 shifts: [8620, 8622, 8624, 39118, 39126, 88124, 88132, 90213, 90215, 90217]
```

Mission 2 - Tamayo's Blood Lab (finite automaton)

```
delta for GATTACAGATTC (rows = states 0..12)
state |   A   C   G   T
    0 |   0   0   1   0
    1 |   2   0   1   0
    2 |   0   0   1   3
    3 |   0   0   1   4
    4 |   5   0   1   0
    5 |   0   6   1   0
    6 |   7   0   1   0
    7 |   0   0   8   0
    8 |   9   0   1   0
    9 |   0   0   1  10
   10 |   0   0   1  11
   11 |   5  12   1   0
   12 |   0   0   1   0

Reverse complement: GAATCTGTAATC

sample_01.txt: forward=0 reverse=0
sample_02.txt: forward=0 reverse=0
sample_03.txt: forward=0 reverse=0
sample_04.txt: forward=0 reverse=0
sample_05.txt: forward=0 reverse=0
sample_06.txt: forward=0 reverse=0
sample_07.txt: forward=0 reverse=3
sample_08.txt: forward=0 reverse=0
sample_09.txt: forward=0 reverse=0
sample_10.txt: forward=0 reverse=0

Demon in disguise: sample_07.txt
```

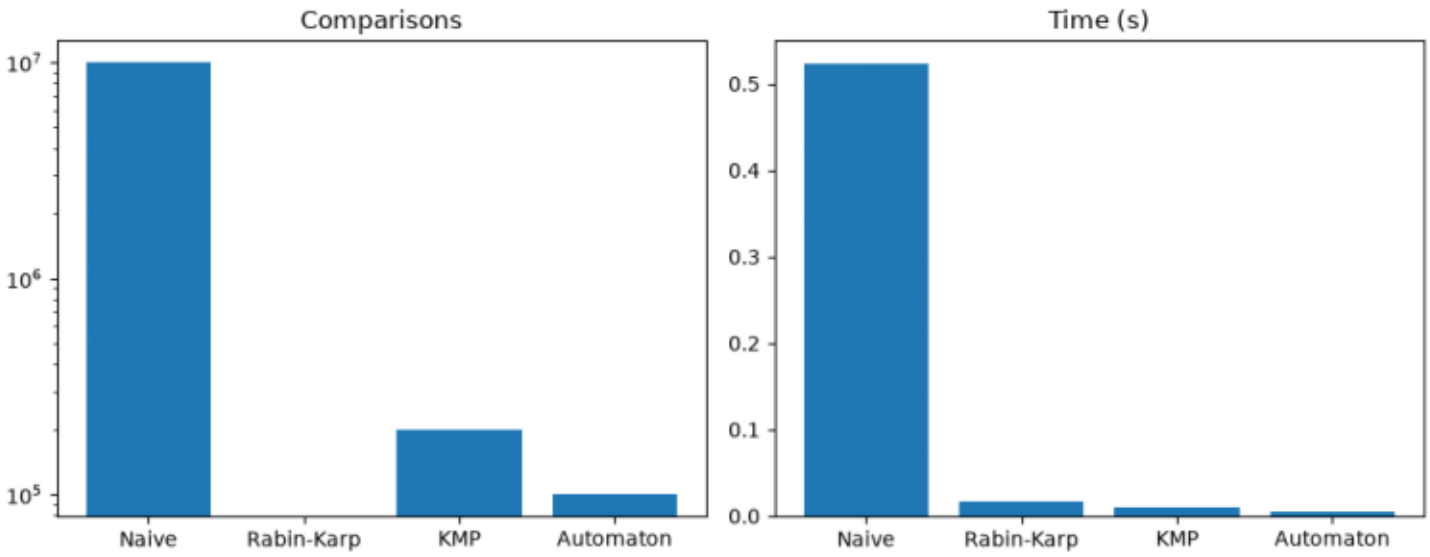
Mission 3 - The Copycat Slayer (Rabin-Karp)

```
Similarity ranking (highest first):
1. Zenitsu   89.00%
2. Kanao     31.86%
3. Inosuke   14.96%
4. Tanjiro    0.00%
5. Genya      0.00%
```

Mission 3: why the word-changer scores lower than deserved

A 40-character window counts only if it matches the manual exactly, so one changed word breaks every window covering it. Rabin-Karp finds exact copies only, not paraphrased ones.

Bonus - Breathing Styles Tournament



Leaderboard

```
Technique      Comparisons    Time (s)
Automaton          100,000      0.0048
KMP                199,901      0.0151
Rabin-Karp               0      0.0192
Naive            9,990,100      0.5371
Saved tournament.png
```

Winner and loser

Winner: the automaton does exactly one lookup per text character (100,000), with no backtracking.
KMP is close: it never moves back in T, so it needs at most 2n comparisons.
Loser: naive restarts at every shift and matches 99 a's before failing, so about n*m comparisons.